

GSLC – Doubly Linked List

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Question

1. Make a comparison between a linked list and a linear array. Which one do you prefer to use and when?
2. Why is a doubly linked list more useful than a singly linked list?
3. Give the advantages and uses of a circular linked list.
4. Specify the use of a header node in a header linked list.
5. Give the linked representation of the following polynomial :
$$7x^3y^2 - 8x^2y + 3xy + 11x - 4$$

Answers

1. Comparison between a linked list and a linear array :
 - a. Linked list :
 - i. A collection of nodes
 - ii. Using node that reference to another node
 - iii. Data is stored dynamically based on the size of the data
 - b. Linear array :
 - i. Value that's stored adjacent to each other in memory
 - ii. Data is stored statically based on user input
 - iii. Using an index to reference the stored data

I prefer to use the Linked list when I frequently insert and delete data in the middle of the collection because it's faster than using a linear array. On the other hand, I prefer to use the linear array when I frequently access the data in the middle by index because then I don't need to traverse the collection to search for the data, I also prefer to use the linear array when the amount of data I need to use is static or doesn't have to change over time like to define a string.

2. A doubly linked list is more useful than a singly linked list because we can traverse the linked list bidirectional. This is especially useful when we need to access a node that's just before the current node, we don't need to traverse the whole linked list again. Instead, we just need to go back 1 time. This is especially useful in erasing a node in the middle of the collection. With a doubly linked list, we can also traverse the linked list In reverse order which may be useful in certain applications.
3. The advantages and the uses of a circular linked list is
 - a. Advantages
 - i. No need to assign NULL in the code unless a node is to be erased.
 - ii. Good for End operation because the head and tail are connected.

- iii. Perform all the regular functions of a singly linked list. And a circular doubly linked list can fully eliminate the need to fully traversal of the linked list.
 - b. Uses
 - i. Can be used to implement various algorithms, such as round-robin scheduling and the Josephus Problem.
 - ii. Can be used in peer-to-peer network to implement a Distributed Hash Table (DHT)
 - iii. Can be used to implement circular queue or circular buffer. Which is a data structure when the newest node is added at the end of the list but the oldest data is removed from the front of the list when it reached a certain size.
- 4. The use of the header in a header linked list is
 - a. Simplify insert and deletion because then the way to delete or insert the first node will be handled the same way as with any other nodes in the list.
 - b. Prevent errors by keeping track of the starting point of the node to traverse the list rather than to keep track of the first node every time it changes.
- 5. The linked list representation of the polynomial is

